CS & IT ENGINEERING

Algorithms

Dynamic Programming

Lecture No.- 01

By- Aditya sir

Recap of Previous Lecture

Topic

Topic

MCST

P49

Topics to be Covered

Topic

Topic Dijkstra SSSP

Topic Intro to DP

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

@ADITYASIR_PW

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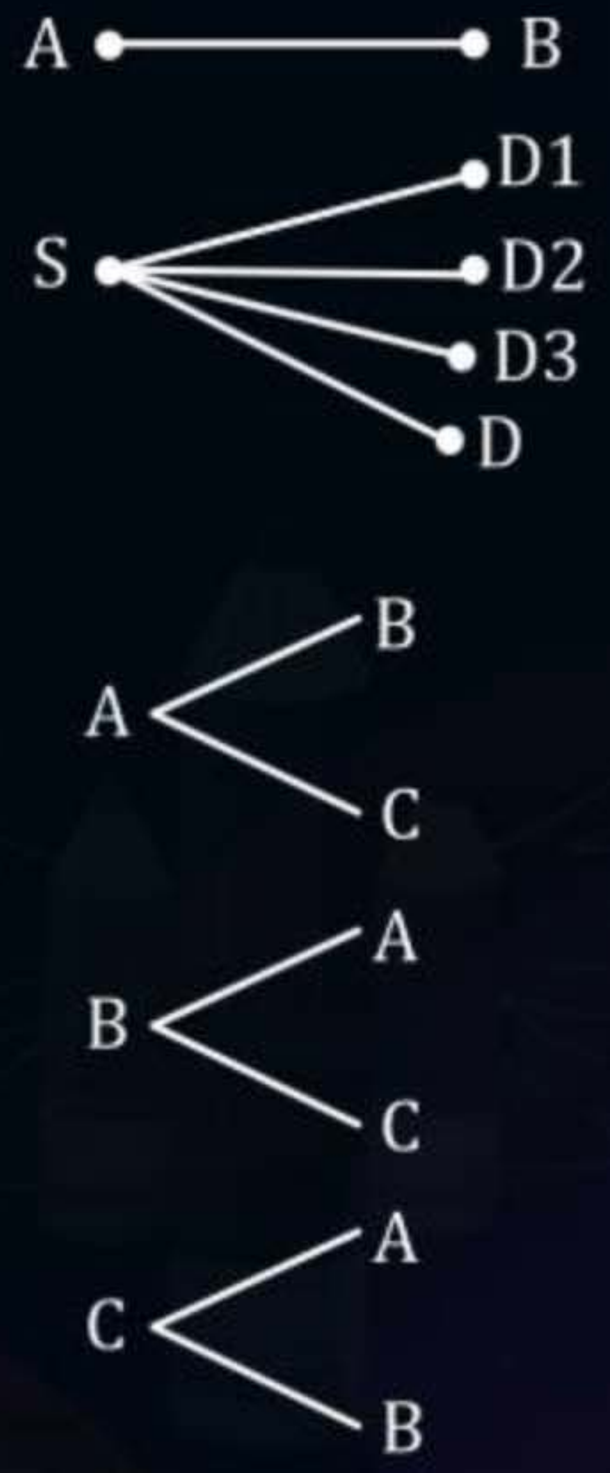
Shortest Path Algorithms Summary

1. Single Pair shortest Path (SPSP):
2. Single source shortest Paths
   (a) Dijkstra's SSSP Alog→ Greedy
   (b) Bellman ford→ Dynamic Programming (DP)
3. All pairs shortest Path:
   (a) Floyd war shall algo
   Dynamic Programming (DP)



#Q. Dijkstra's Single source shortest Path Algorithm (SSSP)

2 Approaches:-

1. Based on Matrix → given only SSSP cost but not paths
2. Based spanning Tree→ Gives both costs as well as paths.

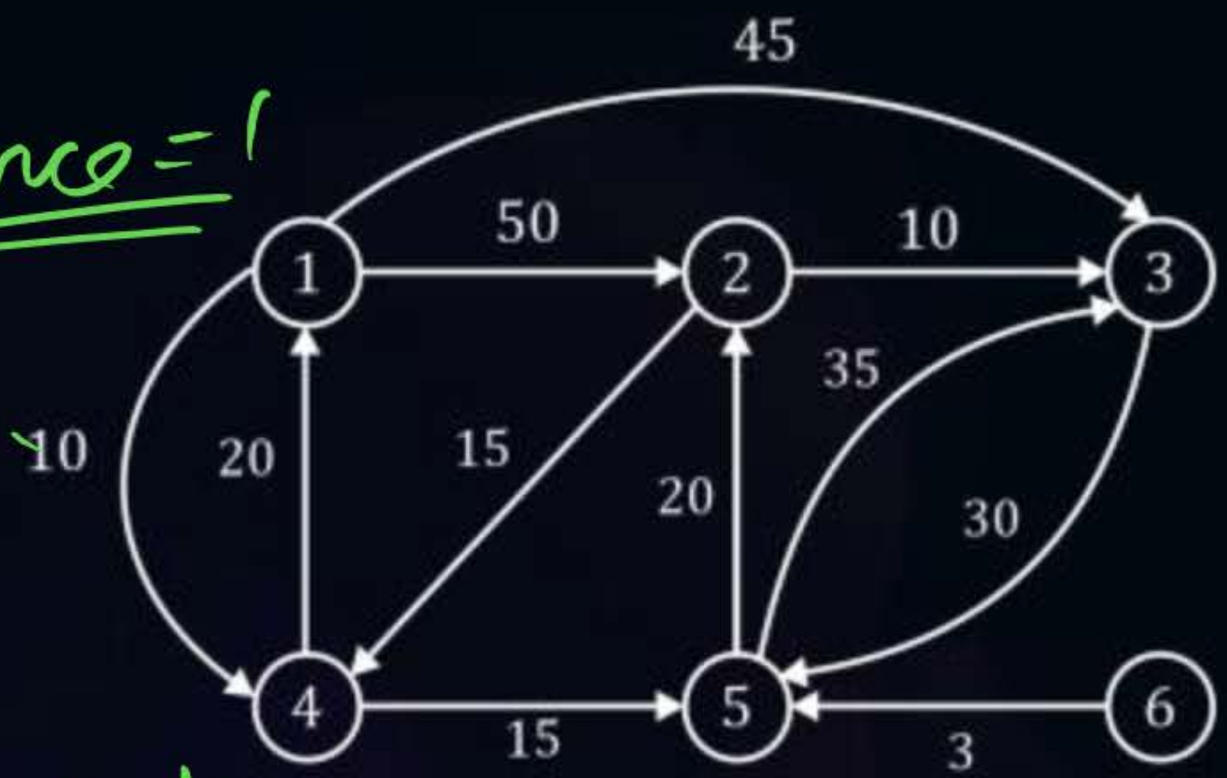
Topic : Greedy Techniques

(Q.2) Source = 6

1. Matrix Based

Given:-

source = 1



Relaxation

Relaxation

vertex Set= {}	1	2	3	4	5	6
S1{1}	-	50	45	10	∞	∞
S2{1,4}	-	50	45	10	25	∞
S3:{1,4,5}	-	45	45	10	25	∞
S4{1,4,5,2}	-	45	45	10	25	∞
S5{1,4,5,2,3}	-	45	45	10	25	∞
S6{1,45,2,3,6}	-	45	45	10	25	∞

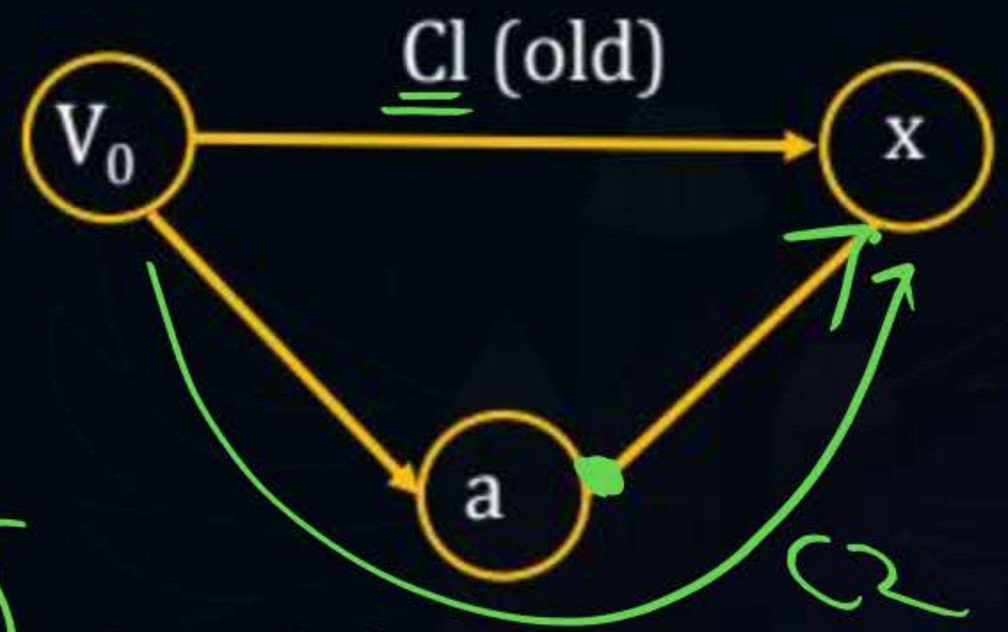
d(n) = distance from source (V0) to a vertex 'x' known so far.

Final answer

$1 \rightarrow 2 : 45$ $\quad$ $1 \rightarrow 4 : 10$

$1 \rightarrow 3 : 45$ $\quad$ $1 \rightarrow 5 : 25$

$1 \rightarrow 6 : \infty$



Relaxation {

Straitly less

If (C2 < C1)

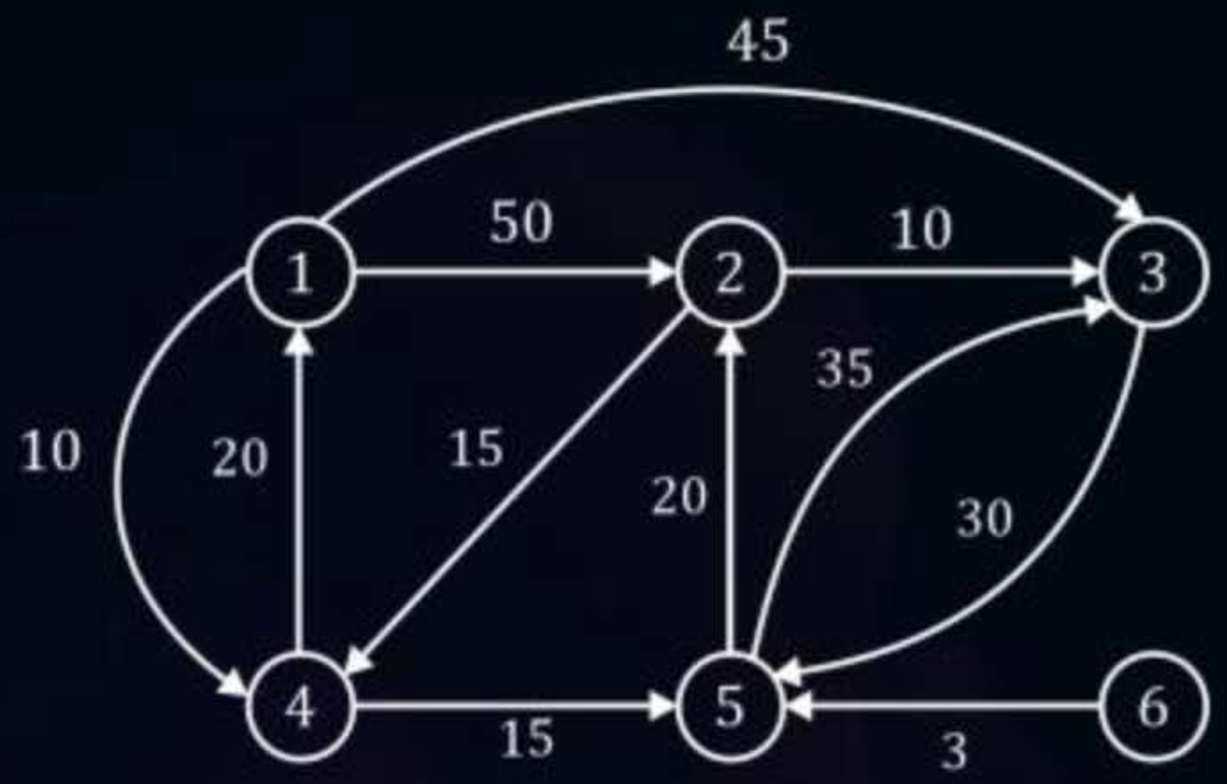
Update d(x) = C2

}

Topic : Greedy Techniques

Practice:- Source

Given:-



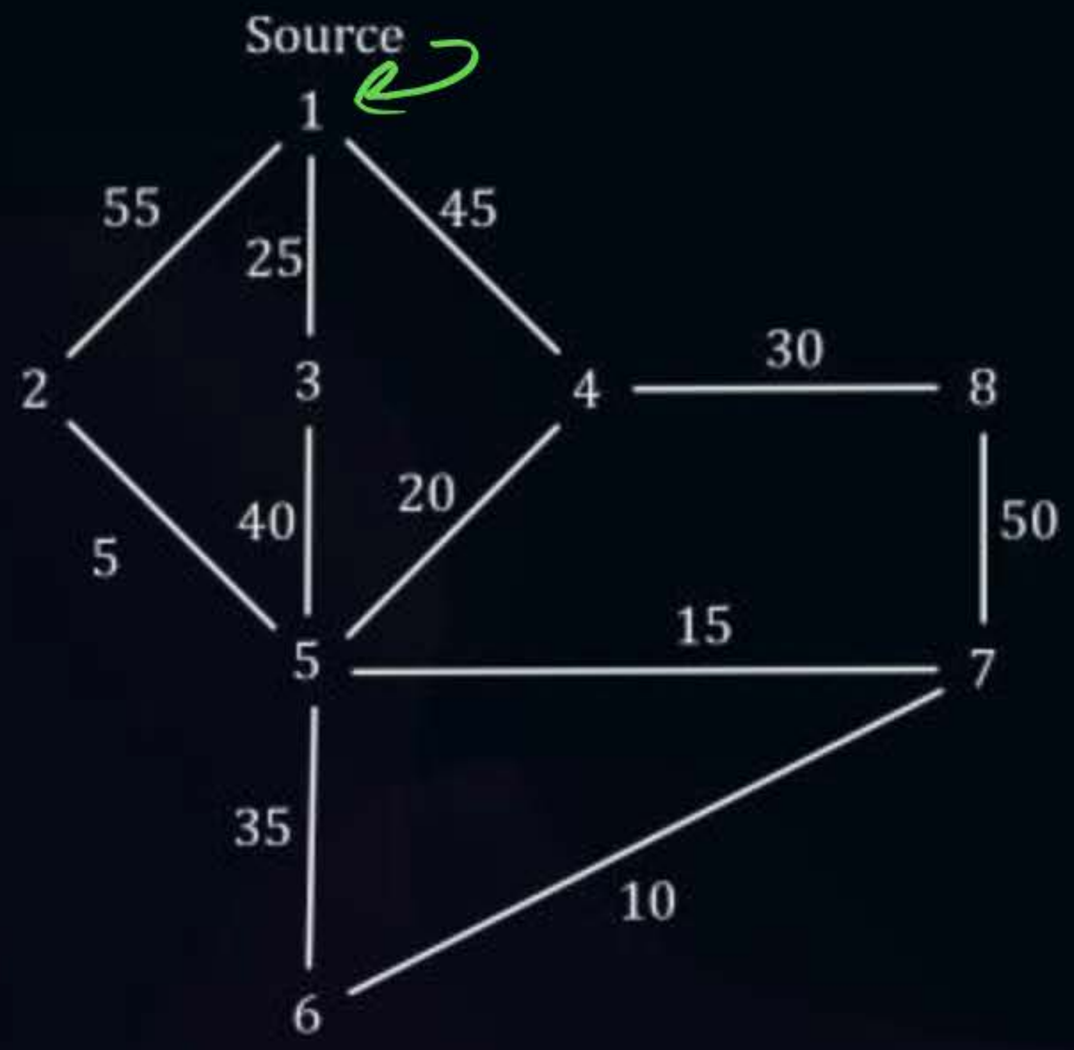
vertex Set= {}	1	2	3	4	5	6
S1{6}	∞	∞	∞	∞	3	-
{6,5}	∞	23	38	∞	3	-
{6,5,2}	∞	23	33	38	3	-
{6,5,2,3}	∞	23	33	38	3	-
{6,5,2,3,4}	58	23	33	38	3	-
{6,5,2,3,4,1}	58	23	33	38	3	-

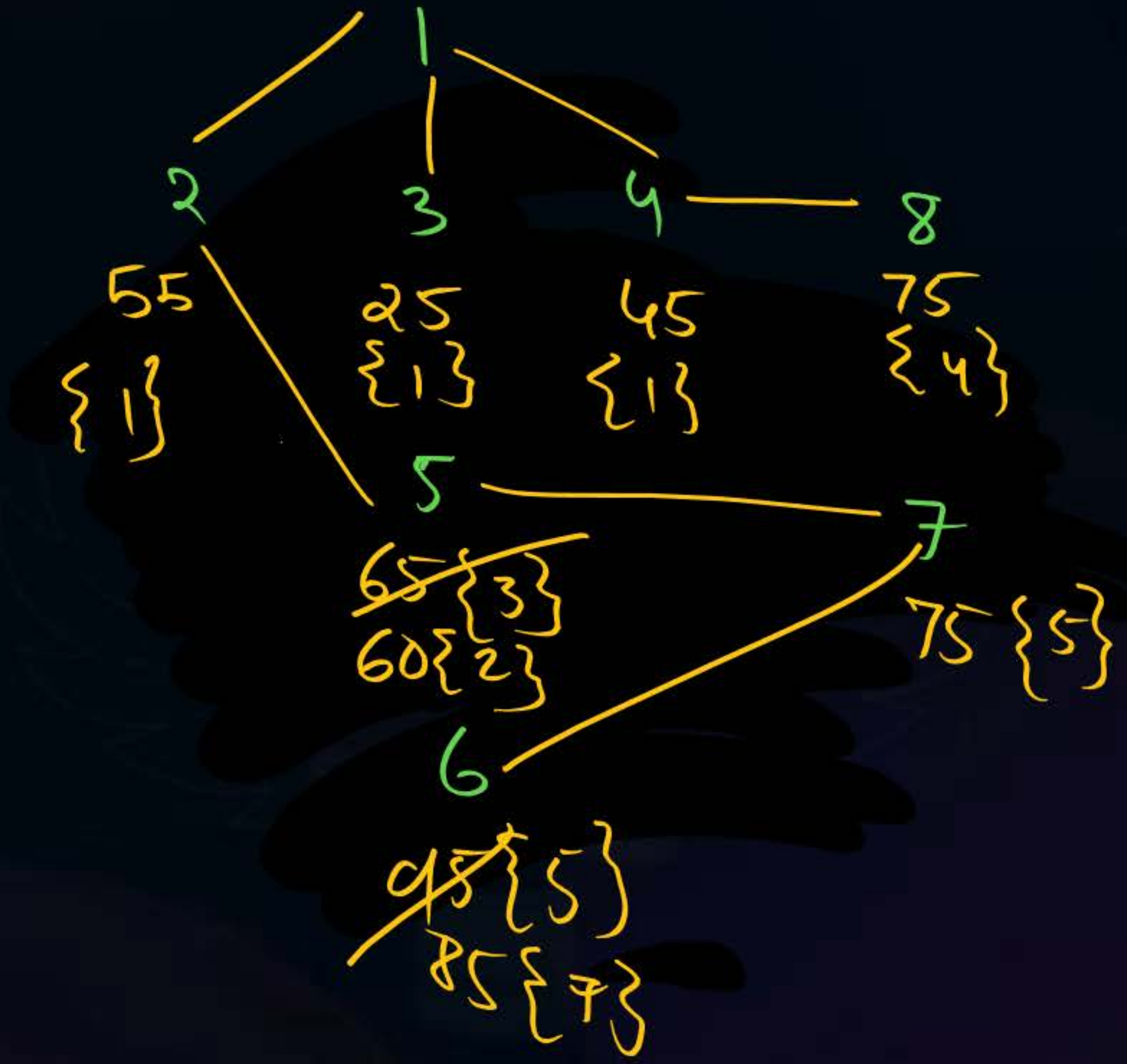
Note :- once a ~~matrix~~ vertex is selected it does not gets furthers relatexed.

Topic : Greedy Techniques

2. Spanning Tree based approach.

Given:





$1 \to 6$?

cost: 85

Path given

By DJ SSSP: $[1 \to 2 \to 5 \to 7 \to 6]$

Topic : Dynamic Programming

```
Algorithm ShortestPaths (v, cos dist,n)
// dist [j], 1 ≤ j ≤ n, is set to the length of the shortest
// path from vertex v to vertex j in a digraph G with n
// vertices. dist[v] is set to zero.- G is represented by its
// cost adjacency matrix cost/.[1: n, 1: n].
    {
    for i := 1 to n do
    {
        // Initialize S.
        S[i] := false; dist[t] := cost[v, i];
  }
    S[v] := true; dist[v] := 0.0; // Put v in S.
    for num := 2 to n - 1 do
```

Topic : Dynamic Programming

```
{
        // Determine n - 1 paths from v.
Choose u from among those vertices not
in S such that dist[u] is minimum;
S[u] := true; // Put u in S.
for (each w adjacent to u with S[w] = false) do
// Update distances.
if (dist [w] > dist[u] + cost[u, w])) then
dist[w]:= dist[u] + cost[u, w];
        }
}
```

Relaxation

Time Complexity analysis:-

1. Non-heap based implementation

   TC→ $O(n^2)$

2. Heap based implementation

   TC→ $(n+e)\log n$

$|V| = n$

$|E| = e$

Topic : Dynamic Programming

MCQ

#Q. Consider a weighted undirected graph with positive edge weights and let uv be an edge ~~in the edge~~ in the graph. It is known that the shortest path from the sources vertex s to u has weight 53 and the shortest path from s to v has weight 65. Which one of the following statements is always true?

54%

Relaxation

A Weight (u, v) < 12

B Weight (u, v) ≤ 12

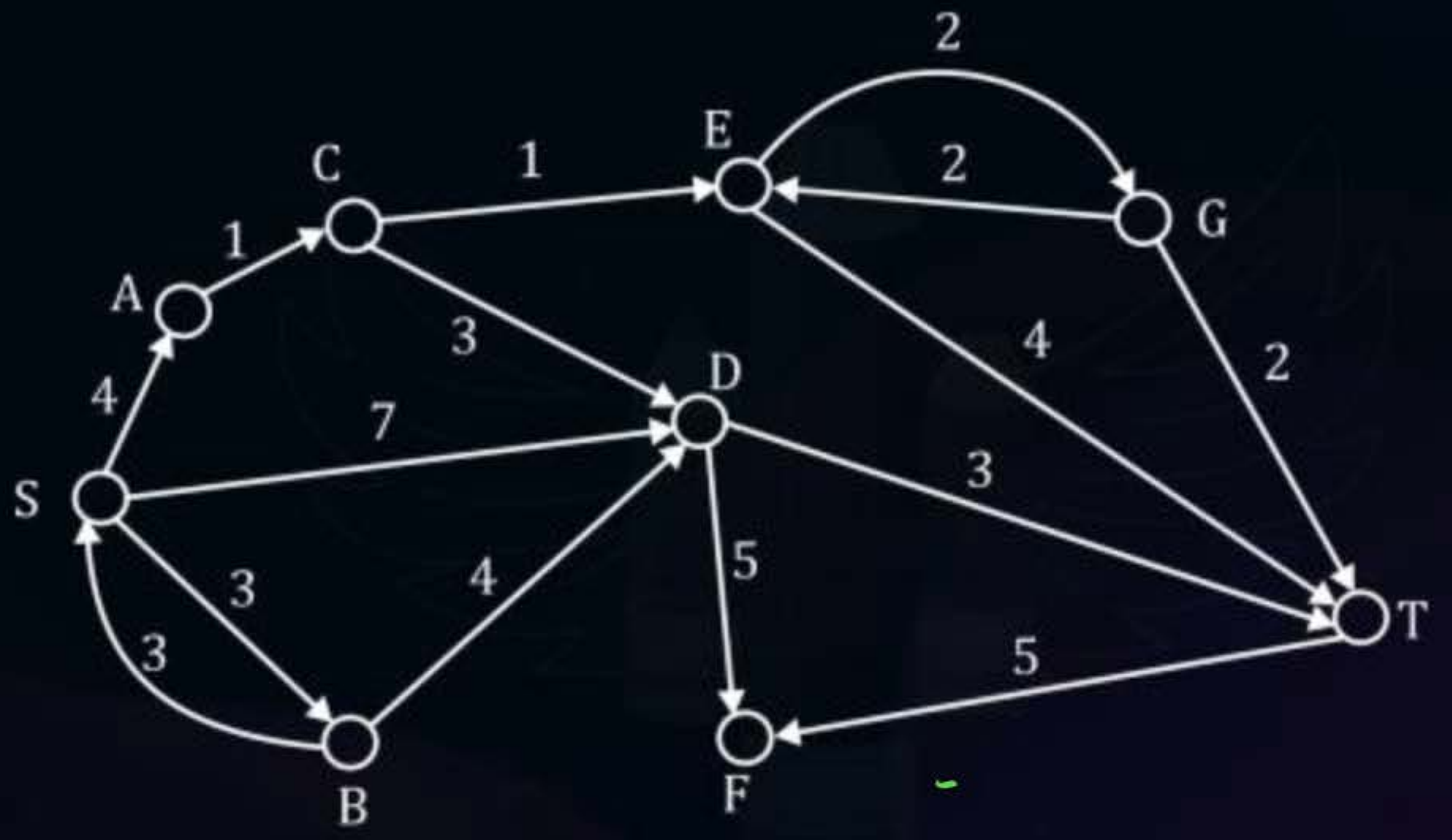
C Weight (u, v) > 12

D Weight (u, v) ≥ 12 ✓

PYQ

#Q. Applying Dijkstra's Algoirthm over the given Graph, which path is reported for 'S' to 'T';

18%



A SBDT → 10

B SDT → 10

C SACEGT → 10

D SACET → 10

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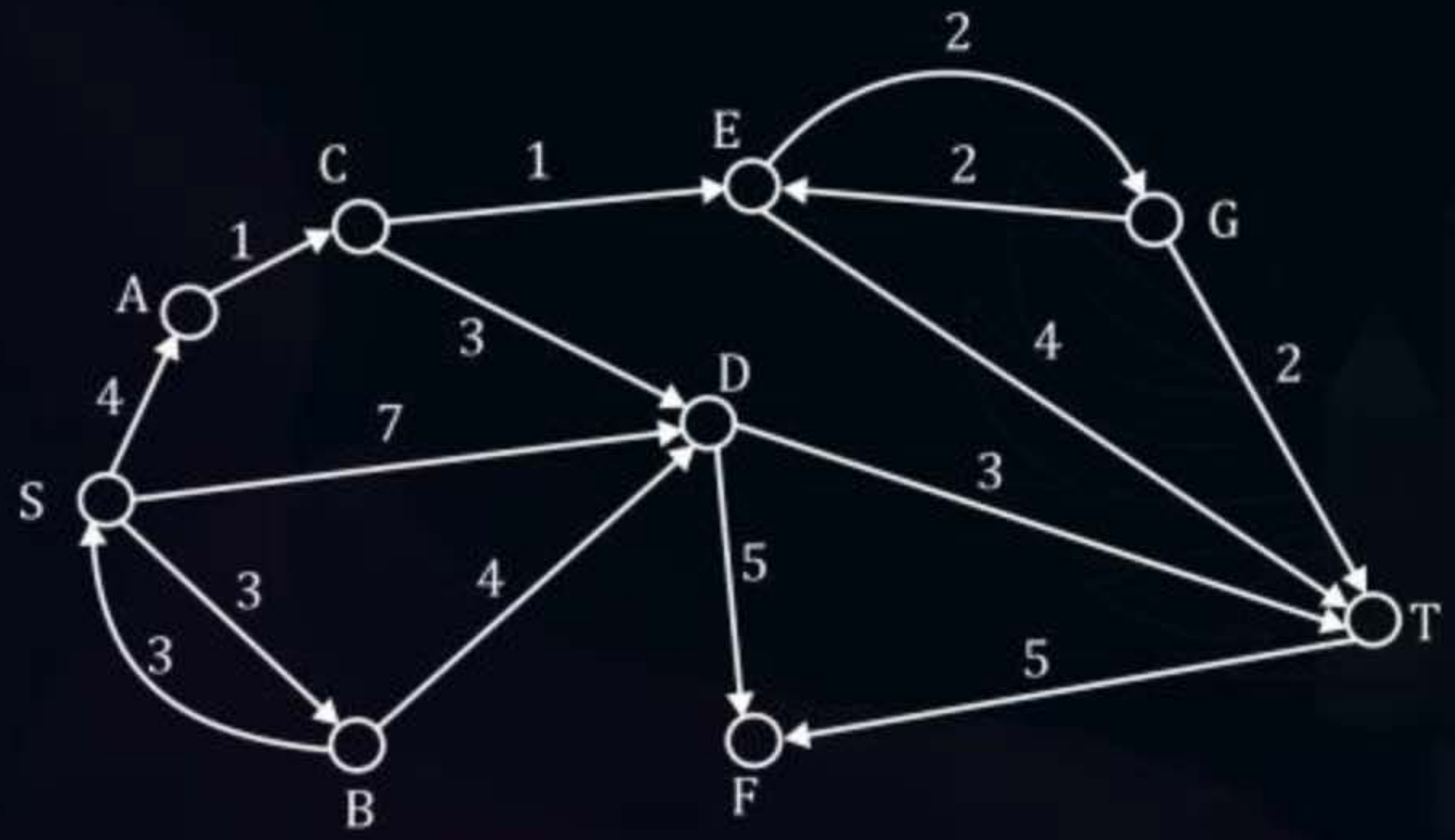
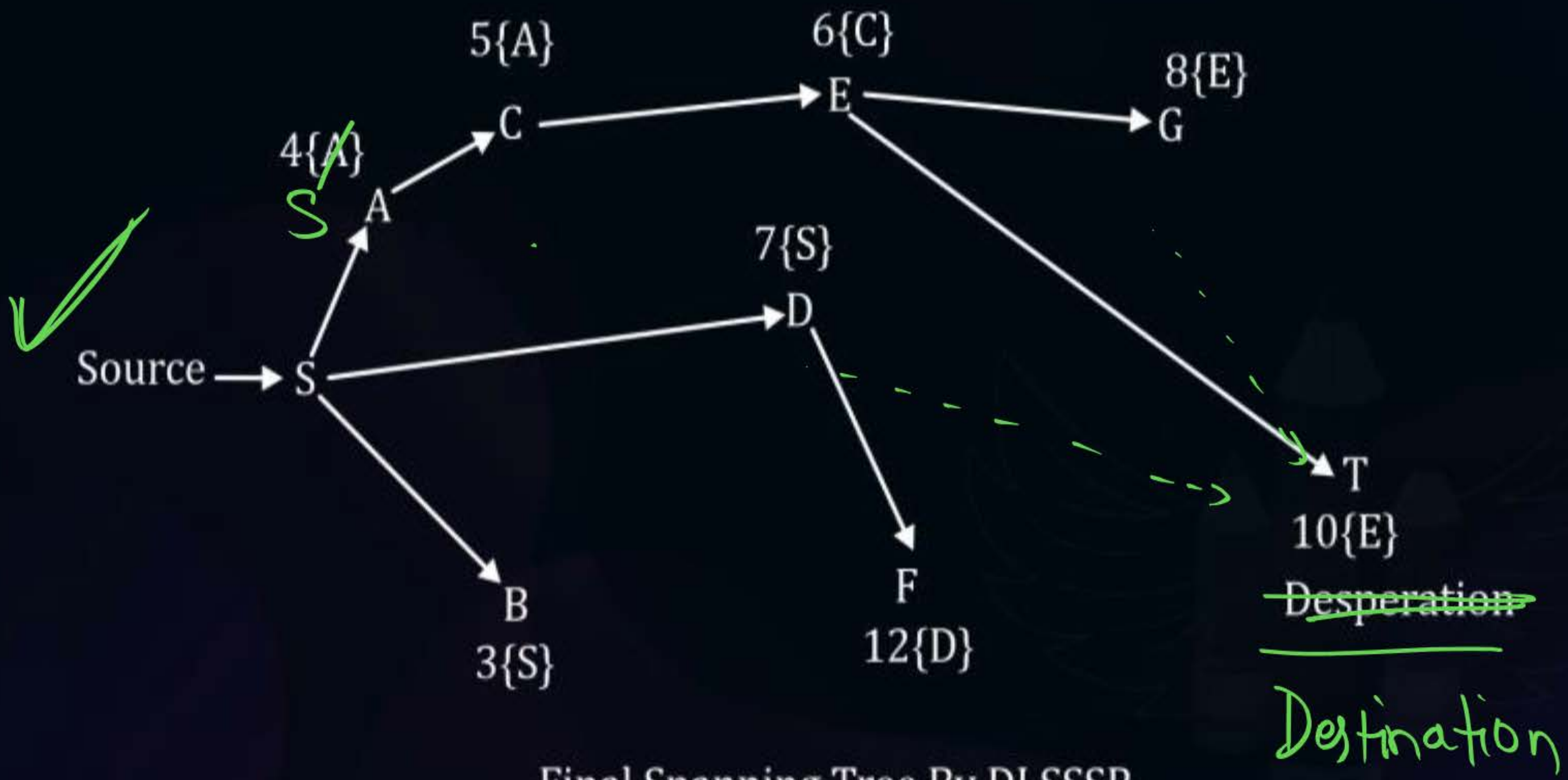
Applying Dijkstra's SSSP Sapnning Tree approach

Spanning: Path reported by DJ SSSP is

Only $S \rightarrow A \rightarrow C \rightarrow E \rightarrow T$





Final Spanning Tree By DJ SSSP.

Relaxation happen only when a strictly lesser path cost is found.

Topic : Dynamic Programming

PYQ

80%

#Q. Let G be weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are true?

T S_1. Minimum spanning Tree of the graph does not change.

S_2. Shortest path between any pair of vertices does not change

33.33%

A Only S_1 true

B Only S_2 true

C Both S_1 and S_2 true

D Both S_1 and S_2 are False

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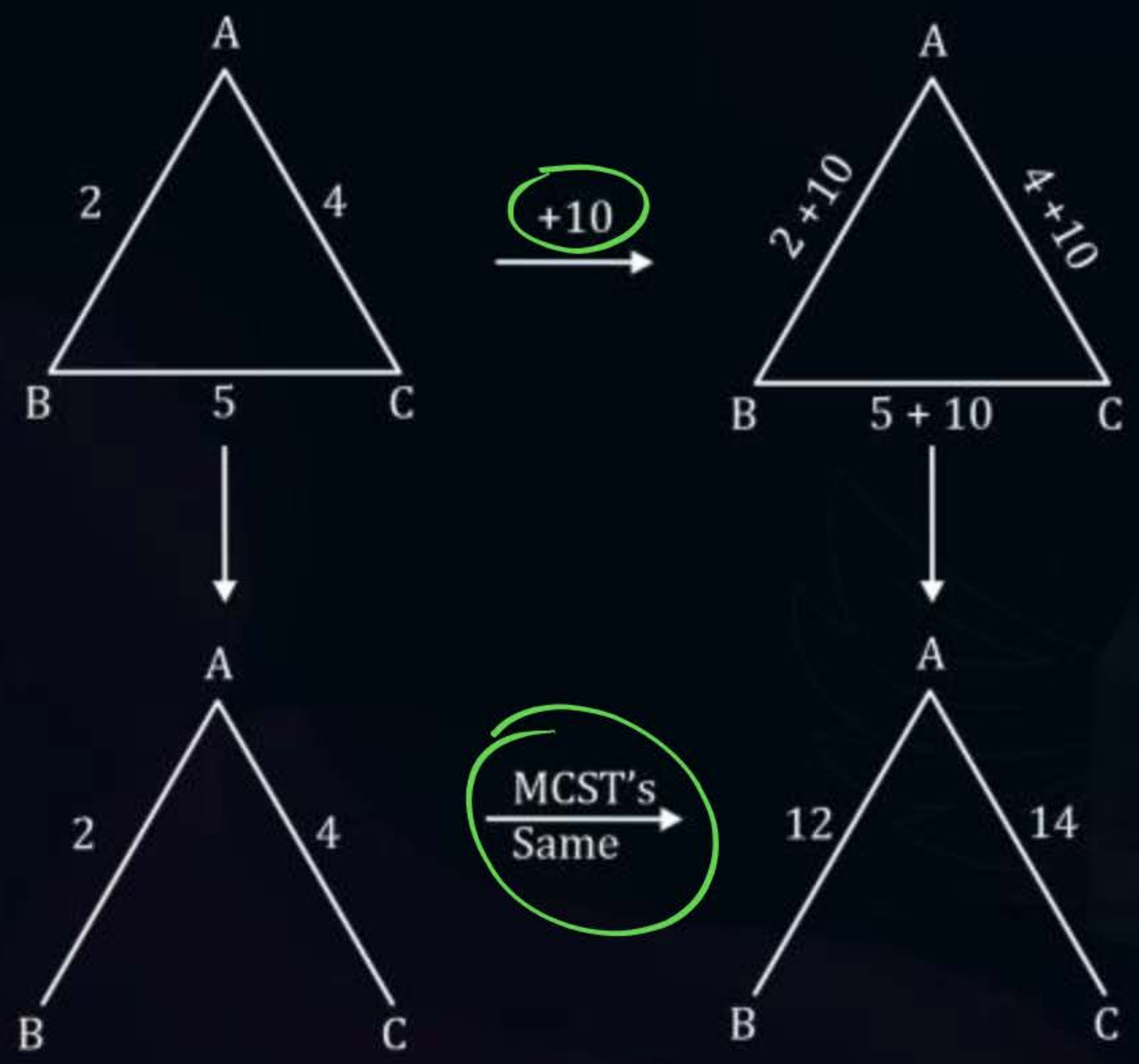
S2:



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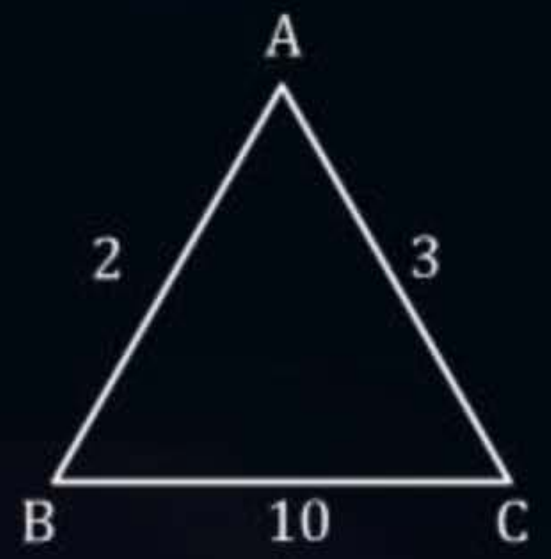
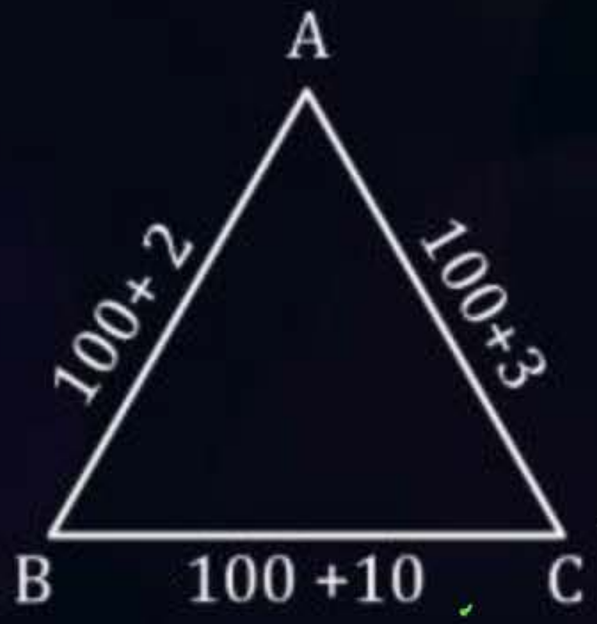
S2:



$\downarrow$ + 100



B→C

Shortest Path from B→ C ?

$B \underset{2}{\rightarrow} A \underset{3}{\rightarrow} C = 2 + 3 = 5$ (BAC)

$\rightarrow$ $B \xrightarrow[110]{} C$

BAC → 205

BC → 110

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80%

#Q. Let G = (V, E) be any connected undirected edge- weighted graph. The weights of the edges in E are positive and distinct. Consider the following statements:

T I. Minimum Spanning Tree of G is always unique

F II. Shortest path between any two vertices of G is always unique.

Which of the above statements is/are necessarily true?

68%

- **A** I only
- **B** II only
- **C** Both I and II
- **D** Neither I and II

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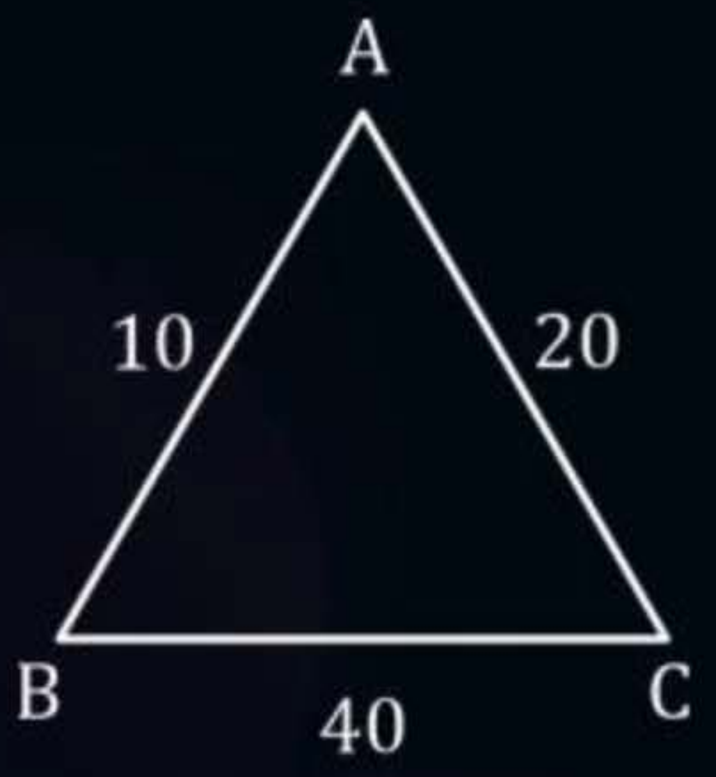
Eg.1.

Unique

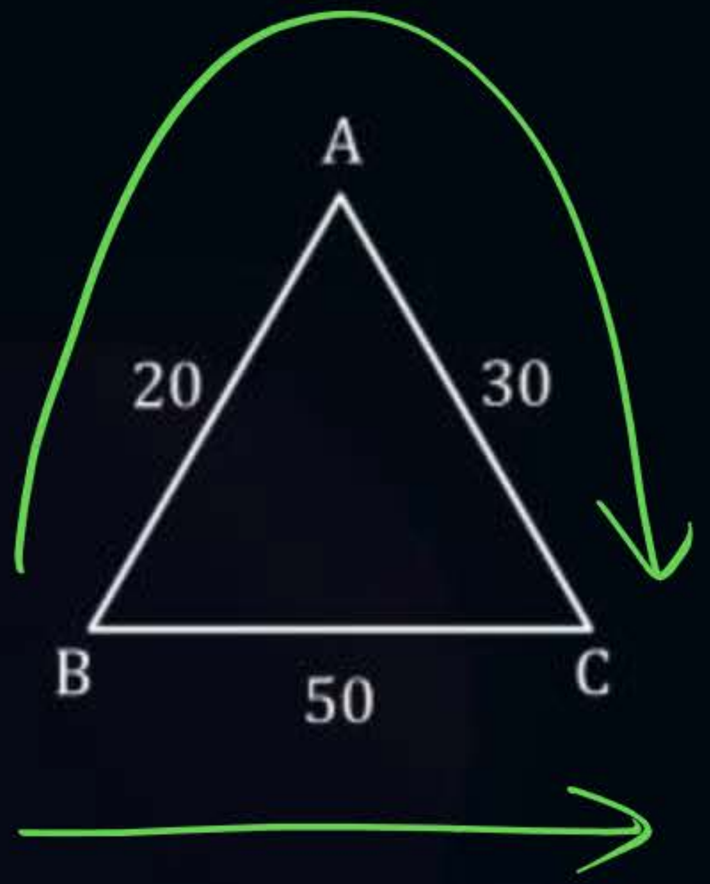


$B \longrightarrow C$

$B \xrightarrow[10]{} A \xrightarrow[20]{} C$

$10 + 20 = 30$

Eg.2.



Shortest Path

$$\boxed{B \longrightarrow C}$$

(1) $B \xrightarrow[20]{} A \xrightarrow[30]{} C$

$20 + 30 = 50$

(2) $B \longrightarrow C = 50$

Shortest Path Is NOT unique in this case.

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HW

#Q. Consider the following table :

Algorithm	Design Paradigms
(P) Kruskal	(i) Divide and Conquer
(Q) Quick sort	(ii) Greedy
(R) Floyd-Warshall	(iii) Dynamic Programming

Match the algorithms to the design paradigms they are based on.

A. (P) $\leftrightarrow$ (ii), (Q) $\leftrightarrow$ (iii), (R) $\leftrightarrow$ (i)

B. (P) $\leftrightarrow$ (iii), (Q) $\leftrightarrow$ (i), (R) $\leftrightarrow$ (ii)

C. (P) $\leftrightarrow$ (ii), (Q) $\leftrightarrow$ (i), (R) $\leftrightarrow$ (iii)

D. (P) $\leftrightarrow$ (i), (Q) $\leftrightarrow$ (ii), (R) $\leftrightarrow$ (iii)

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Introduction to Dynamic Programming:

~~Sorting~~/ Tabulating the values or Results of the sub- Problems.

Storing

$$\begin{tikzcd}
 & \text{Problems} \arrow[dl] \arrow[dr] & \\
\text{Decision} \arrow[d] & & \text{Optimization} \\
\text{SOS} & &
\end{tikzcd}$$

Topic : Dynamic Programming

V. V. Imp

Dynamic Programming (DP) is an algorithm design method used for solving problems, whose solutions are viewed as a result of making a set / sequence of decisions.

- One way of making these decisions is to make them one at a time in a step-wise (sequential) step-by-step manner and never make an erroneous decision. This is true of all problems solvable by Greedy method.)
- For many other problems it is not possible to make step-wise decisions based on local information available at every step, In such a manner that the sequence of decision made is optimal.

V. imp

eg:- 0/1 Knapsack

eg

optimal : 60

greedy : 44

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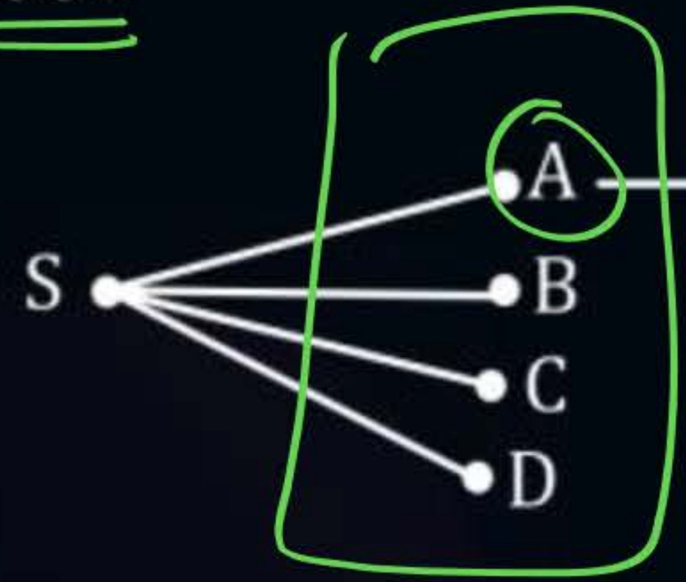
Eg. 0/1 Knapsack



Main diff. between greedy & DP.

Sometimes greedy appr. Does not give optimal solution to some problems.

2 mins Summary

Topic

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Topic

Topic

Dijkstra SSSP

Intro to DP

THANK - YOU